

Smart Hospital appointment & Queue Management Business Analysis Case Study



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INTRODUCTION

Project Background

A large hospital currently manages appointments, patient records, prescriptions, billing, and inventory using separate systems and manual processes. This causes delays, duplicate data, and poor patient experience.

The Smart Hospital Management System aims to integrate all hospital operations into a centralized digital platform.

Objective

To analyze existing business problems and recommend improvements using Business Analysis techniques.

PROBLEM STATEMENT

- Long patient waiting times
- Manual appointment scheduling
- Lost patient records
- Delayed billing process
- Poor communication between departments
- Difficulty tracking medicine inventory

SWOT ANALYSIS

Strengths	Weaknesses
Skilled medical staff	Manual processes
Large patient base	Data duplication
Established reputation	Lack of integration

Opportunities	Threats
Digital healthcare growth	Cybersecurity risks
AI-assisted diagnosis	Competitive hospitals
Telemedicine services	Regulatory changes

Analysis Conclusion

The hospital possesses strong medical expertise but requires digital transformation to improve efficiency and patient satisfaction.

GAP ANALYSIS

Current State

- Paper-based records
- Manual billing
- Separate systems
- Appointment queues

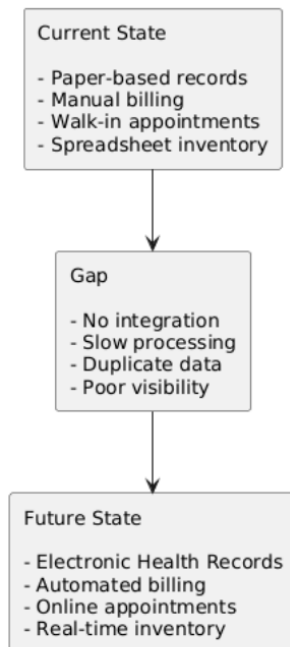
Future State

- Electronic Health Records (EHR)
- Automated billing
- Integrated hospital platform
- Online appointments

GAP TABLE

Area	Current	Future	Gap
Records	Paper files	Digital records	High
Billing	Manual	Automated	Medium
Appointments	Walk-ins	Online booking	High
Inventory	Spreadsheet	Real-time tracking	High

GAP Analysis - Smart Hospital System



ROOT CASE ANALYSIS

Problem

Long patient waiting times

Root Causes

- Manual registration
- Limited appointment visibility
- Duplicate paperwork
- Department communication delays

Conclusion

Waiting times are primarily caused by inefficient patient registration and scheduling processes.

5 Whys ANALYSIS

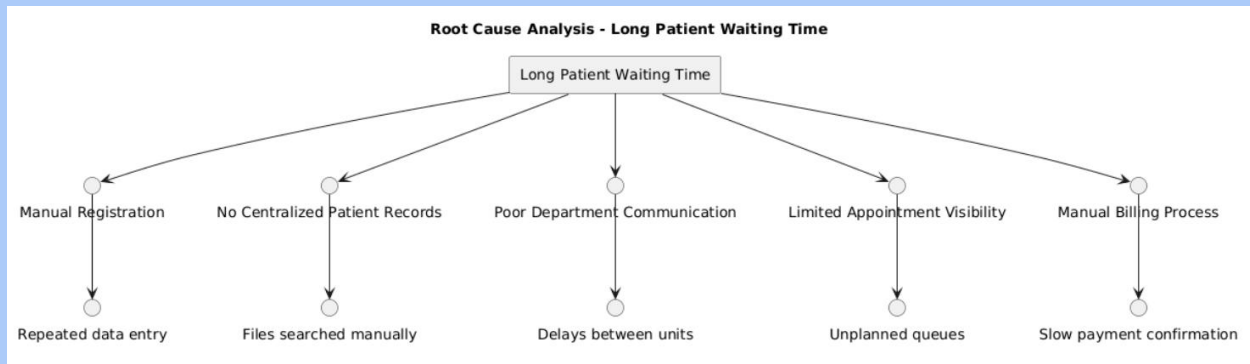
Problem:

Patients wait over 2 hours before consultation.

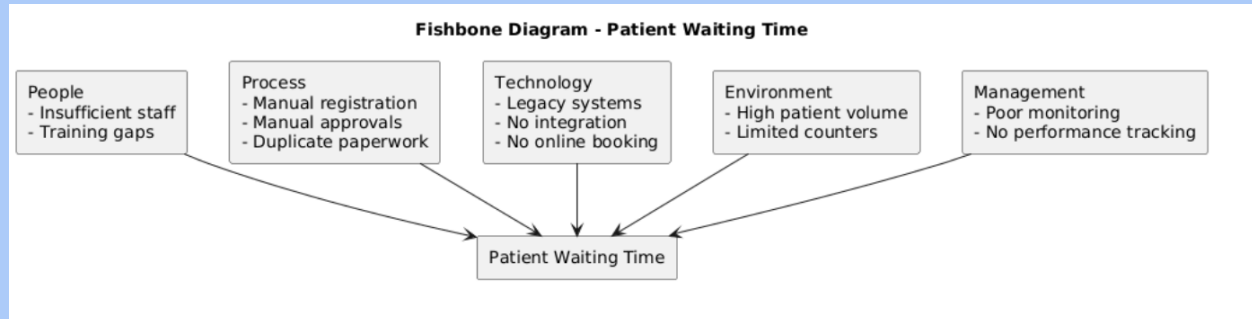
Why	Answer
Why 1	Registration takes too long
Why 2	Patient data entered manually
Why 3	No centralized system
Why 4	Departments use separate databases
Why 5	No integrated hospital platform

Root Cause

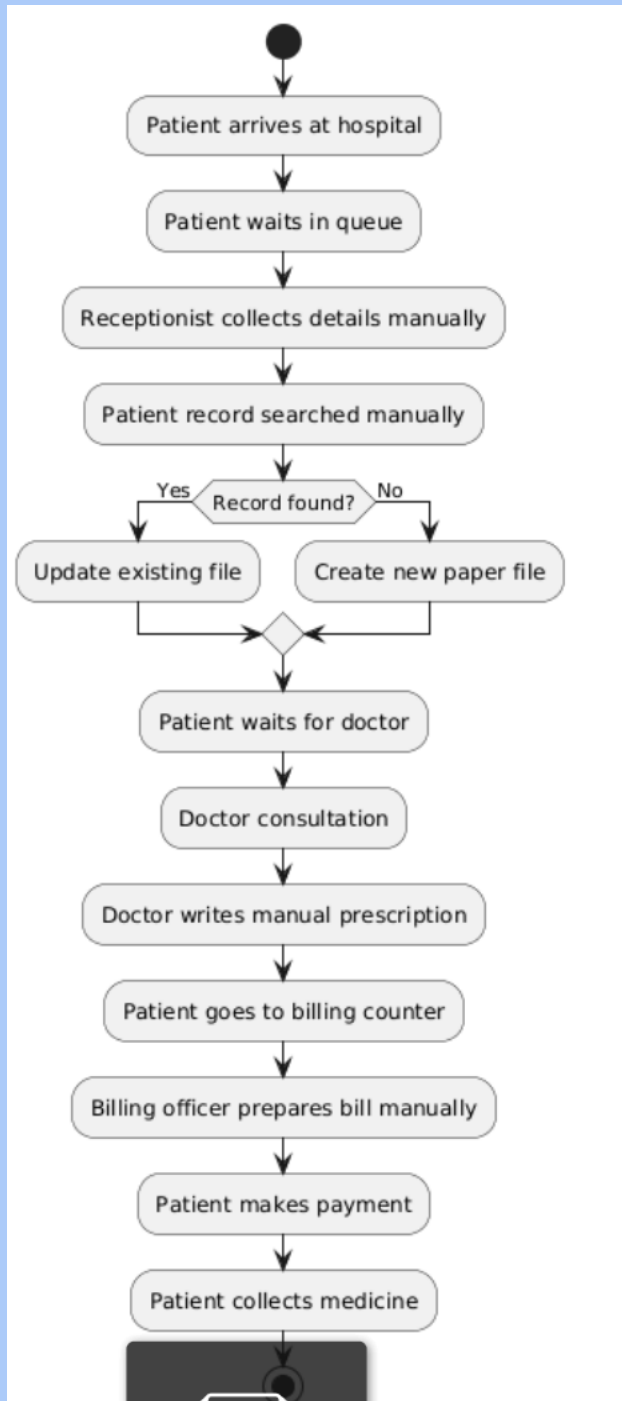
Lack of an integrated digital hospital management system.



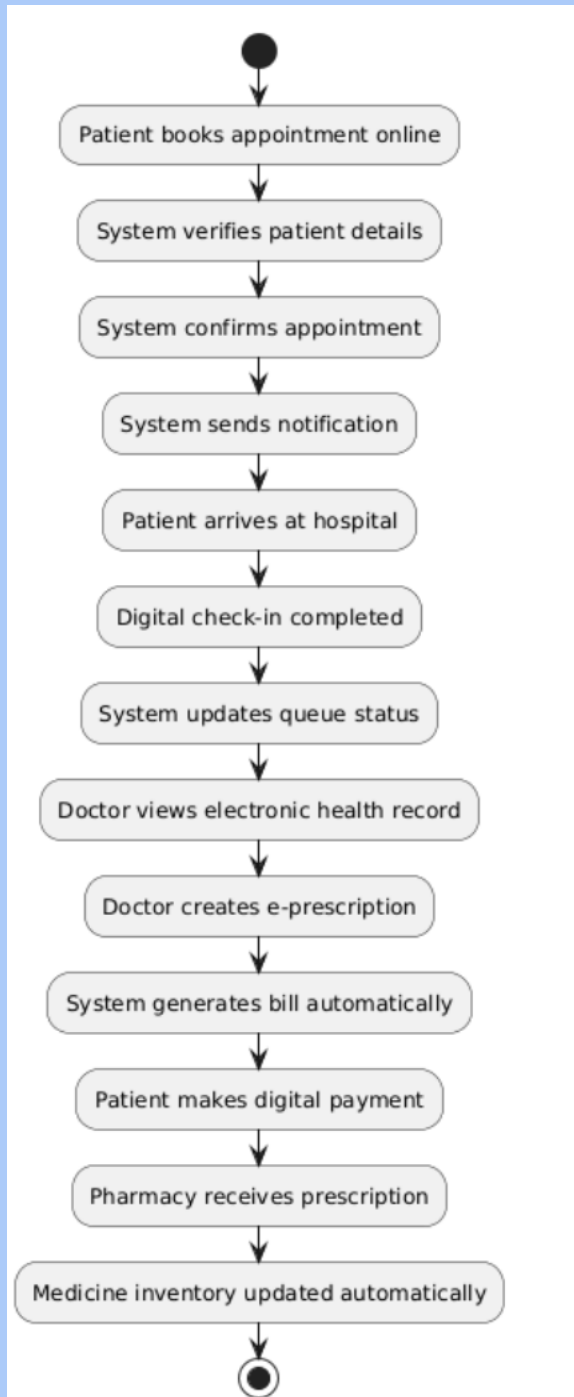
Fishbone Diagram Analysis



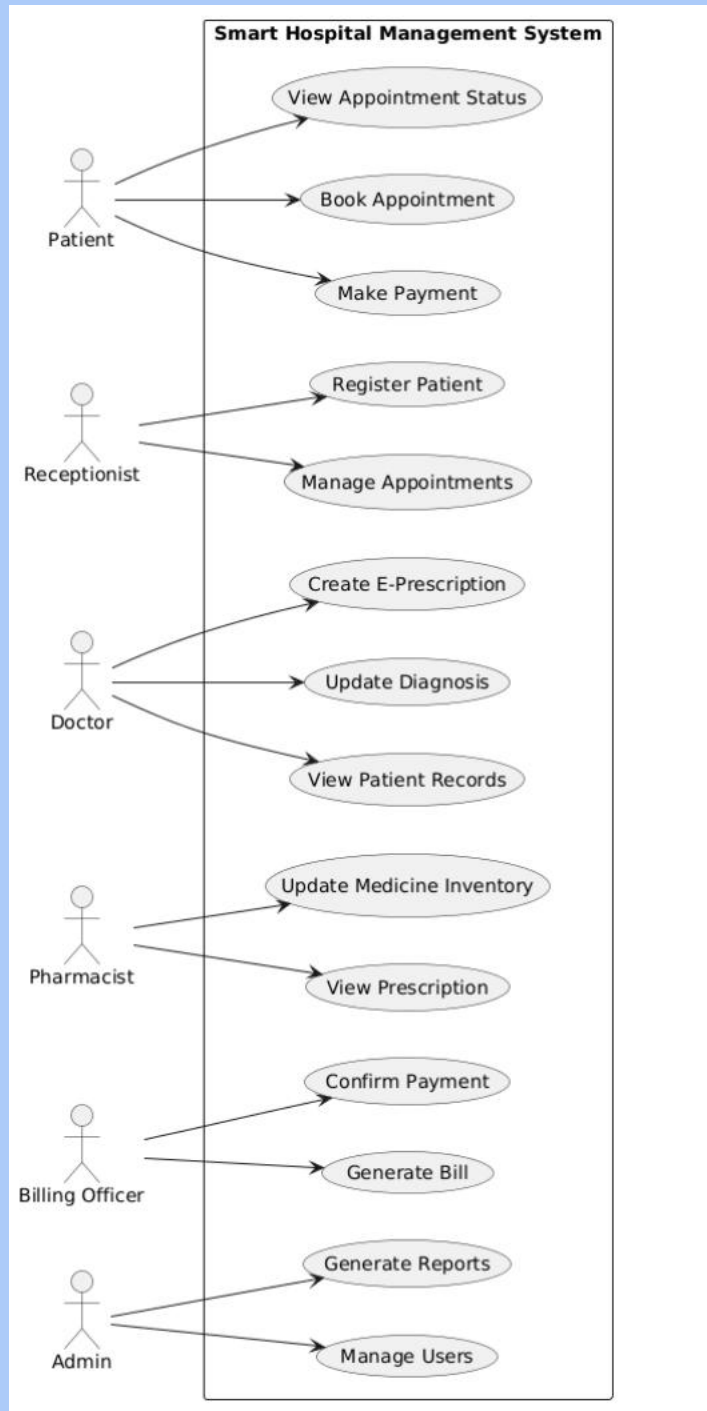
AS-IS Process Flow



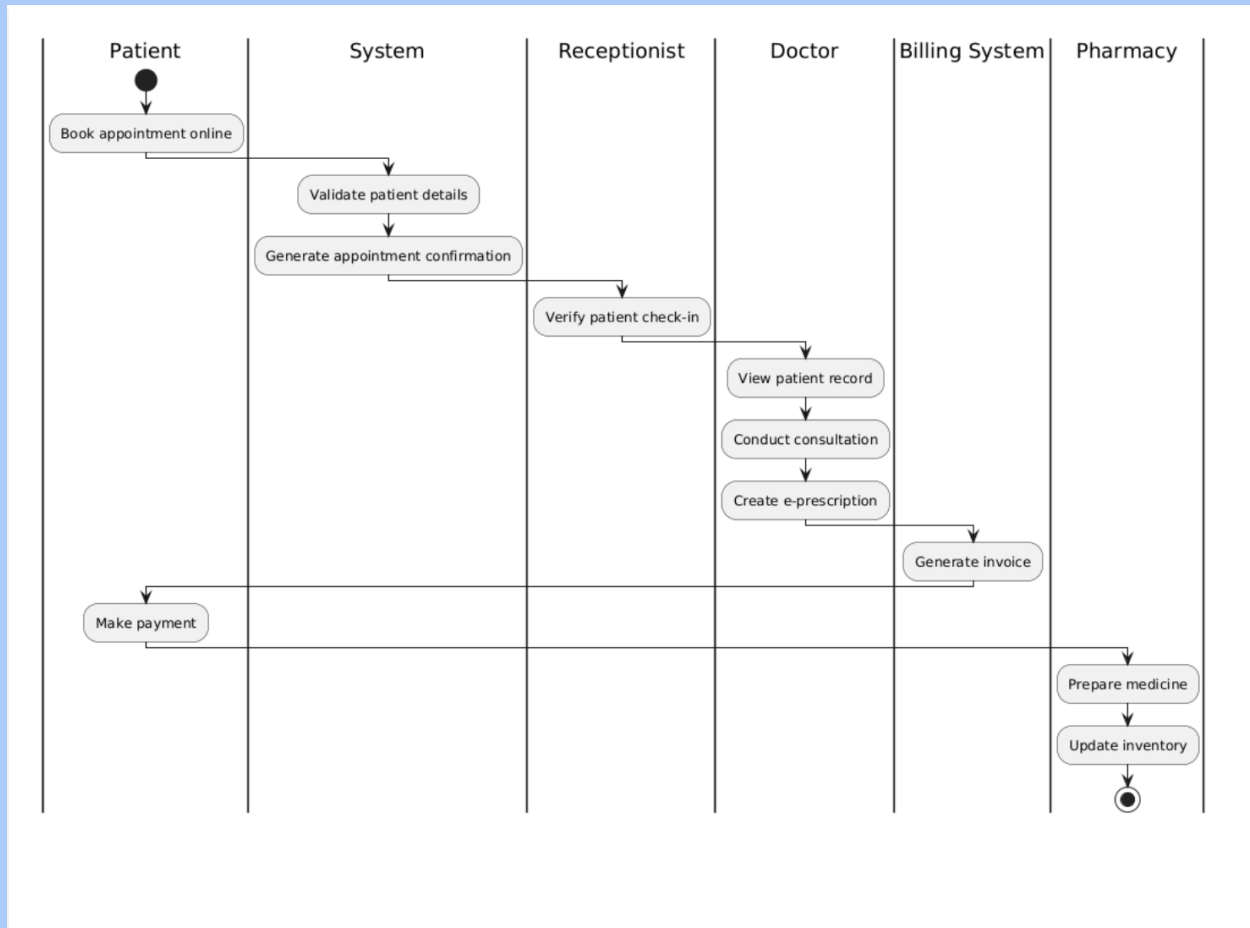
TO-BE Process Flow



Use Case Diagram



BPMN Style Swimlane Diagram



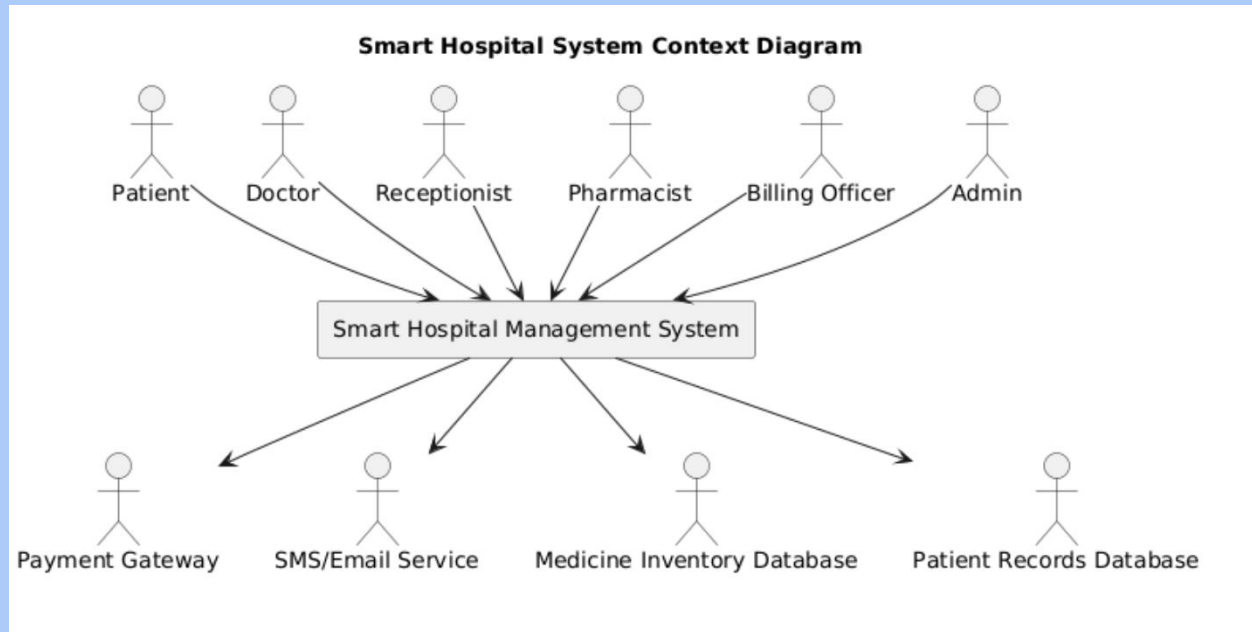
MoSCoW Prioritization

Requirement	Priority
Online Appointment Booking	Must Have
Electronic Health Records	Must Have
Automated Billing	Must Have
SMS Notifications	Should Have
Mobile App	Could Have
AI Health Prediction	Won't Have (Phase 1)

Conclusion

Core hospital operations are prioritized first, while advanced features are deferred to later phases.

System Context Diagram



Recommendations

1. Implement Electronic Health Records.
2. Introducing Online Appointment Booking.
3. Automate Billing Process.
4. Deploy Real-Time Inventory Tracking.
5. Provide Staff Training.
6. Establish Data Security Controls.

Business Value

Expected Benefits:

- 50% reduction in waiting time
- Improved patient satisfaction
- Better operational efficiency
- Reduced paperwork
- Faster billing process
- Improved inventory visibility

Conclusion

The Smart Hospital Management System analysis identified several operational inefficiencies caused by manual processes and fragmented systems. Through SWOT Analysis, GAP Analysis, Root Cause Analysis, BPMN, Process Mapping, Fishbone Diagram, 5 Whys, and MoSCoW Prioritization, key improvement areas were identified. Implementing the proposed digital solutions will significantly enhance hospital efficiency, patient experience, and overall service quality.